Review Sheet 2: Managing Variability

1. A Petroleum Refinery*

An integrated petroleum company is considering expansion of its one unloading facility at its Australian refinery. Due to random variations in weather, loading delays, and other factors, ships arriving at the refinery to unload crude oil arrive according to the Poisson distribution (CV_a=1) with an average arrival rate of 8 ships per week. Service time is exponential with an average service rate of 12 ships per week (CV_S=1)

a) Current performance measures:

i. What is the average number of ships waiting to gain access to the single unloading facility?

ii. What is the average time a ship must wait before beginning to deliver its cargo to the refinery?

iii. What is the average total time (waiting plus actual delivery) that a ship spends at the refinery?

b) The company has under consideration a second unloading berth that could be rented for $6,000 per week. The service time for this berth would also be exponential (CV_s=1) with the same service rate of 12 ships per week as the company's own berth. The opportunity cost of a ship not being at sea is $30,000 per week.

i. If the second berth is rented, what will be the average number of ships waiting?

ii. What would be the average time a ship would wait?

iii. Is the benefit of reduced waiting time (in dollars) worth the rental cost for the second berth?

c) An alternative way to improve unloading facilities is for the company to modify the current operations to speed up the unloading process. Specifically, the new service time would still be exponential (CV_s=1), but with a mean service rate of 15 ships per week. The incremental cost of the new operations would be $ 4,000 per week.

i. Compute the (new) average number of ships waiting.

ii. Compute the (new) expected waiting time and total time for ships.

iii. Is the benefit worth the incremental cost?

iv. How would you answer i - iii above if, instead of speeding up the service rate of the facility, the new operations made the service time constant, but still with the old service rate of 12 ships per week?

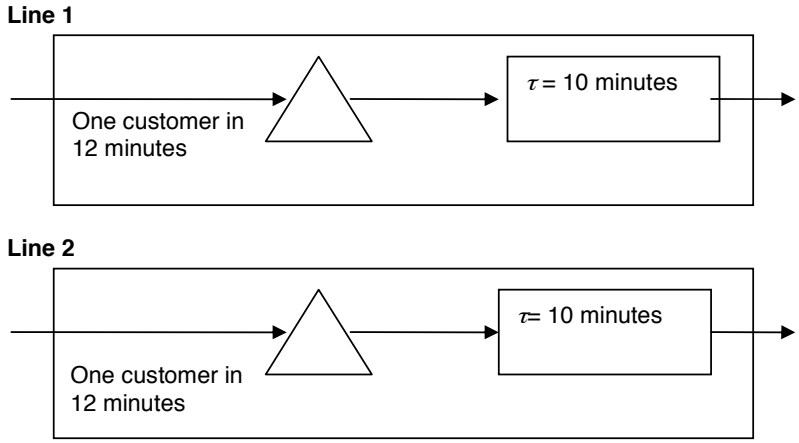
2. Queuing Theory in Court (Based on an actual case)

A worker at the State Unemployment Office is responsible for processing a company's forms when it opens for business. The worker can process an average of 4 forms per week. In 1992 an average of 1.8 companies per week submitted forms for processing, and the worker had an order flow time of 0.45 weeks. In 1993 an average of 3.9 companies per week submitted forms for processing and the worker had a 5 week order flow time. The poor worker was fired, but later sued to get her job back. The court said that because the amount of work submitted to the worker had approximately doubled, the worker's backlog also should have doubled. Because her backlog increased by more than a factor of 10, she must have been slacking off, so the state was justified in firing her. Assume exponential service times and Poisson arrivals and a FCFS (first come, first served) queueing discipline.

Use the concepts and tools learned in POM to defend the worker.

3. Service Facility Design

A service organization has the following process design. There are two separate but identical service centers (with one employee each) that serve two distinct geographic areas. Each center has random service time with a mean of 10 minutes and a standard deviation of 10 minutes. Customers arrive at each center with an inter-arrival time of 12 minutes. The standard d eviation of this inter-arrival time is 12 minutes.

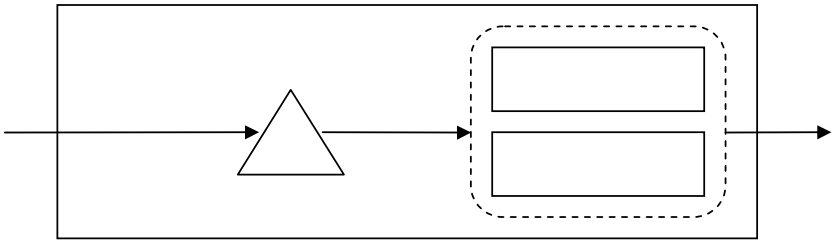


a) ***What is the current average wait time in minutes?****

b) ***The organization is considering strategies to reduce wait times. In particular, it is considering reducing service time from 10 minutes to 5 minutes. In this case, the standard deviation of service time would also be reduced from 10 minutes to 5 minutes. What is the average wait time in minutes if this strategy is implemented?****

c) ***Another strategy to reduce waiting time involves standardizing the activity, which would reduce the standard deviation of the service time from 10 to 2.5 minutes. Service time of 10 minutes will remain unchanged. What is the average wait time if this strategy is implemented?****

d) ***The final strategy considered is to merge the two centers as shown below (in this case, the standard deviation of the inter-arrival times will be 6 minutes). What is the average wait time if this strategy is implemented?****



Appendix 1: L_q Values for the Multi-server Queue

Values of L_q for s servers, with mean utilisation rate ρ, assuming Poisson arrivals and exponential service times (known as a $M/M/s$ queue).

Utilisation rate	*Number of servers* (*s*)				
(ρ)*	1	2	3	4	5
.10	.0111	.0020	.0004	.0001	.0000
.20	.0500	.0167	.0062	.0024	.0010
.30	.1286	.0593	.0300	.0159	.0086
.35	.1885	.0977	.0552	.0325	.0196
.40	.2667	.1524	.0941	.0605	.0398
.45	.3682	.2285	.1522	.1052	.0743
.50	.5000	.3333	.2368	.1739	.1304
.55	.6722	.4771	.3583	.2772	.2185
.60	.9000	.6750	.5321	.4306	.3542
.62	1.0116	.7743	.6213	.5109	.4269
.64	1.1378	.8880	.7246	.6051	.5130
.66	1.2812	1.0188	.8446	.7158	.6152
.68	1.4450	1.1698	.9847	.8461	.7367
.70	1.6333	1.3451	1.1488	1.0002	.8816
.72	1.8514	1.5500	1.3423	1.1834	1.0553
.74	2.1062	1.7914	1.5721	1.4025	1.2646
.76	2.4067	2.0785	1.8472	1.6668	1.5187
.78	2.7655	2.4237	2.1803	1.9887	1.8302
.80	3.2000	2.8444	2.5888	2.3857	2.2165
.82	3.7356	3.3661	3.0979	2.8832	2.7029
.84	4.4100	4.0265	3.7456	3.5190	3.3273
.86	5.2829	4.8852	4.5914	4.3526	4.1493
.88	6.4533	6.0414	5.7345	5.3834	5.2682
.90	8.1000	7.6737	7.3535	7.0898	6.8624
.92	10.5800	10.1392	9.8056	9.5290	9.2893
.94	14.7267	14.2712	13.9240	13.6344	13.3821
.96	23.0400	22.5698	22.2088	21.9060	21.6408
.98	48.0200	47.5350	47.1602	46.8439	46.5656
.99	98.0101	97.5176	97.1357	96.8127	96.4274

* When one is faced with a value of ρ that lies between two values in the table, one may have to interpolate between the two values of Lq associated with the two successive values of ρ. For example, consider $s = 2$ with ρ = 0.75. This utilization rate lies in between the two table values of 0.74 and 0.76. The Lq for this M/M/2 queue can be approximated by the following linear interpolation:

$$L_q = 1.7914 + \frac{(2.0785 - 1.7914)}{(0.76 - 0.74)}(0.75 - 0.74) = 1.7914 + 0.1435 = 1.9349$$